

AP CSP | Lesson 30: Responsible Computing

Homework | Multiple-choice and written practice

Name: _____ Class: _____ Date: _____

Time: 40 minutes | Total: 30 teacher marks (not an AP score) | Show working.

Original AP-style MCQs and classroom written practice; these are not official AP questions.

Complete all 12 original questions (30 marks, about 40 minutes). Q5 requires exactly TWO choices. Use the stated fictional scenarios, models and simplified permission cards. Explain each decision with evidence; never enter or submit real passwords or personal details.

1. A figure's card grants reading only. A team wants to copy it into a public map. Which conclusion is supported? [1 mark]

- A. Online access grants copying permission. B. Get copying permission or replace it; credit alone does not grant permission.
C. Downloading the figure transfers its ownership. D. No watermark means copying is allowed.

Answer: _____

2. Which statement most accurately describes open-source software? [1 mark]

- A. Any program that is free of charge is open source. B. Any visible code has no owner or license conditions.
C. Code is open source only if nobody is allowed to modify it. D. Its license permits uses such as modification and redistribution; making code visible alone is insufficient.

Answer: _____

3. Original Card A allows sharing and adapting an icon for any purpose, with credit and notice of changes. A team recolors it. Which action follows the card? [1 mark]

- A. Credit the creator and state that the colors changed. B. Use it with no credit because the colors changed.
C. Credit the creator but claim the team made no changes. D. Treat permission to adapt as permission to claim sole authorship.

Answer: _____

4. Which claim about open access is most accurate for this course? [1 mark]

- A. It means research has no author. B. It gives ownership of every research figure to every reader.
C. It broadens access to research and many uses, while authorship and applicable terms still matter. D. It means any free-to-read page grants unlimited reuse.

Answer: _____

5. Select exactly TWO supported statements. [1 mark]

- A. A source citation automatically authorizes every intended use. B. Permitted reuse can still raise ethical concerns about harm.
C. Creative Commons licenses all impose exactly the same conditions. D. Record the creator and follow the credit and change-notice conditions.

Answer: _____

6. In a synthetic model, network and device support makes 85 of 100 residents eligible to contribute. What follows? [1 mark]

- A. Exactly 85 residents will certainly submit reports. B. 85 are eligible in this model, but fewer may actually participate.
C. The intervention guarantees that the sample will be unbiased. D. All 100 residents now have no barriers.

Answer: _____

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Written responses | Questions 7-9

7. [4 marks]

A fictional team copies a reading-only research figure into its public map, adds the author's name, and says, "Giving credit means we can use it however we want." Explain permission, attribution, the flaw in the team's claim, and one appropriate next action.

8. [4 marks]

Original classroom Card A: "Shade icon by Map Art Team; source identifier A-01. Share and adapt for any purpose, with credit and an indication of changes." The team resizes the icon and changes its colors. State whether this use can follow the card. Write a credit record identifying the material, creator, source identifier and both changes.

9. [4 marks]

Original classroom cards: B grants reading only for a research figure; C is the team's self-created code, allowed for this project with its author recorded; D is a photograph with no license information. The team wants all three in its public map. Give an appropriate action for each, and explain why "it is online" does not establish permission.

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Written responses | Questions 10-12

10. [4 marks]

The fictional community has 100 people: 40 with no barrier; 20 network only; 15 device only; 10 skills only; 5 accessible-UI only; and 10 with both network and device barriers. Costs: network 2, device 2, skills 1, accessible UI 1. With budget 4, analyze choosing network plus device. Show cost and eligible count, identify unresolved barriers, and explain one limit of the count.

11. [4 marks]

Within-group reported need for more shade is North 20% and South 80%. The sample is North 80% / South 20%; the population is North 60% / South 40%. Calculate the sample estimate and population-weighted reference, compare them in percentage points, and explain one bias that matching zone shares may leave unresolved.

12. [4 marks]

The map only needs North/South summaries of shade ratings. Recommend a field set and explain it, recommend a release format and explain its benefit, choose one suitable authentication or encryption measure and explain its role, and explain a remaining risk. Use no real personal details.
